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26SP//MAT102A

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Intro Statistics

Final Project Proposal

Answers using **BLUE** font.

1. Read through the variable list in the Final Project and select a quantitative variable you would like to center your analysis on. List the exact variable name below.

Quantitative Variable 1:

1. Next, you will need to select another quantitative variable from the list that may have an association with this variable. List the exact variable name below.

Quantitative Variable 2: Pets

1. Next, you will need a qualitative variable with two groups to compare to do an analysis on a difference in means. This variable will be Gender for this dataset.

Qualitative Variable 1: Gender

Two groups: 1) Males 2) Females

1. Next pick a second categorical/qualitative variable from the list that you would like to analyze a proportion on. List the exact variable name below and **give the option of the variable that you are choosing to find the proportion.**

Categorical/Qualitative Variable 2: Allergies Option: Yes

1. You will need to see if there is a relationship between your two quantitative variables by creating a scatterplot and regression line.

Investigative Question	**Variables Involved**	**Type of Analysis**

Is there an association between the number of siblings and the number of pets among students in this sample?	Siblings and Pets	Scatterplot, Correlation, and Regression Equation

1. Next, you will need to analyze a mean using your quantitative variable 1.

Investigative Question	**Variable Involved**	**Type of Analysis**
Describe the distribution of _siblings_ of the students in this sample.	Siblings	Boxplot/Histogram
Estimate the average _number of siblings_ of all students.		CI for a mean
Is the average _number of siblings_ different National Average Average_? American family has approximately 1.9 children. Source: https://www.census.gov/topics/families/families-and-households.html		Test for a mean (Optional)

1. Next, you will need to use your categorical variable 2 to analyze proportions.

Investigative Question	**Variable Involved**	**Type of Analysis**
In this sample, how does _allergies_ differ amongst the students?	allergies	Bar chart/Pie chart
Estimate the proportion of all students who _have allergies_.		CI for a proportion
Is the proportion of all students who _have allergies_ different than _National Average_? CDC National Center for Health Statistics, 31.7% of U.S. adults have at least one diagnosed allergy. Source: https://www.cdc.gov/nchs/fastats/allergies.htm		Test for a proportion (Optional)

1. Now combine your categorical variable 1 and your quantitative variable 1 to analyze a difference in means.

Investigative Question	**Variables Involved**	**Type of Analysis**

In this sample, how does the _number of siblings_ vary between males and females?	Side-by-Side Box plot
Estimate the difference in mean _number of siblings_ between males and females.	CI for a difference in means
Is there a difference in mean _number of siblings_ between males and females?	Test for a difference in means

Research Sources

Proportion with allergies $p_0 = 0.317$
https://www.cdc.gov/nchs/fastats/allergies.htm

Average pets per household $\mu_0 = 2.6$
https://rainwalkpetinsurance.com/pet-ownership-statistics/